

Prerequisite

Differentiation



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Limits

Limit

- If $x \rightarrow a, f(x) \rightarrow L$, then L is called limit of the function $f(x)$.
- $\lim_{x \rightarrow a} f(x) = L$
- Left Hand Limit of f at $x = a$ is $\lim_{x \rightarrow a^-} f(x)$
- Right Hand Limit of f at $x = a$ is $\lim_{x \rightarrow a^+} f(x)$
- If the Right and Left Hand Limits coincide
The limit of $f(x)$ at $x = a$ and $\lim_{x \rightarrow a} f(x)$

Algebra of Limits

- $\lim_{x \rightarrow a} [f(x) \pm g(x)] = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x)$
- $\lim_{x \rightarrow a} [f(x) \times g(x)] = \lim_{x \rightarrow a} f(x) \times \lim_{x \rightarrow a} g(x)$
- $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$
whenever $\lim_{x \rightarrow a} g(x) \neq 0$

Limits of Trigonometric Functions

- $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$
- $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$
- $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$

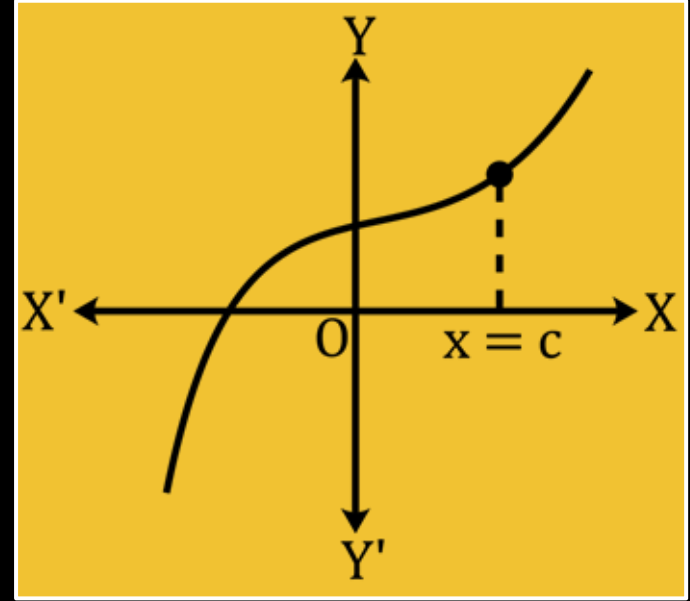
Continuity

Let f be a real function on a subset of the real numbers and let c be a point in the domain of f . Then f is continuous at c if

$$\lim_{x \rightarrow c} f(x) = f(c)$$

$$\lim_{x \rightarrow c^+} f(x) = \lim_{x \rightarrow c^-} f(x) = f(c)$$

If f is not continuous at $c \Rightarrow f$ is discontinuous at c and c is called a point of discontinuity of f .

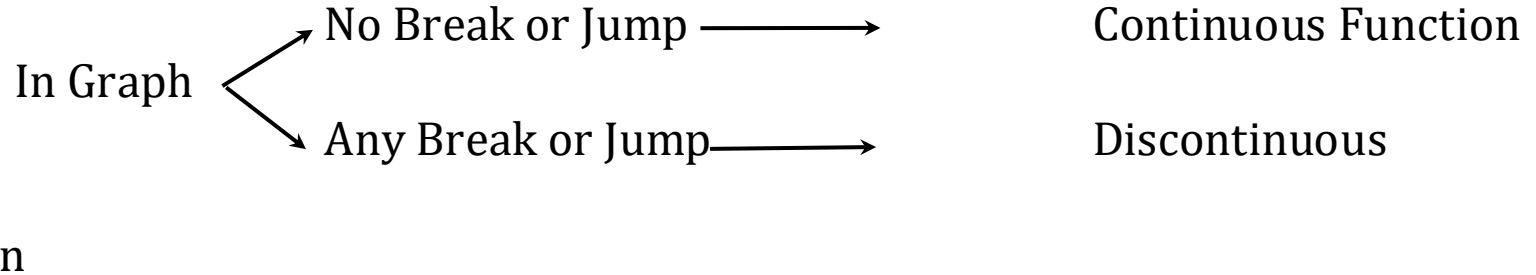


1. Continuity of Function at point $x = c$

$$\lim_{x \rightarrow c} f(x) = f(c)$$

$$\lim_{x \rightarrow c^+} f(x) = \lim_{x \rightarrow c^-} f(x) = f(c)$$

1. Geometrical Meaning of Continuity.



Differentiability and Derivatives of Composite Functions

Example: $y = f(x)$

$$\text{Slope of CA} = \frac{f(a-h)-f(a)}{a-h-a} = \frac{f(a-h)-f(a)}{-h}$$

as $h \rightarrow 0 \Rightarrow C \rightarrow A$

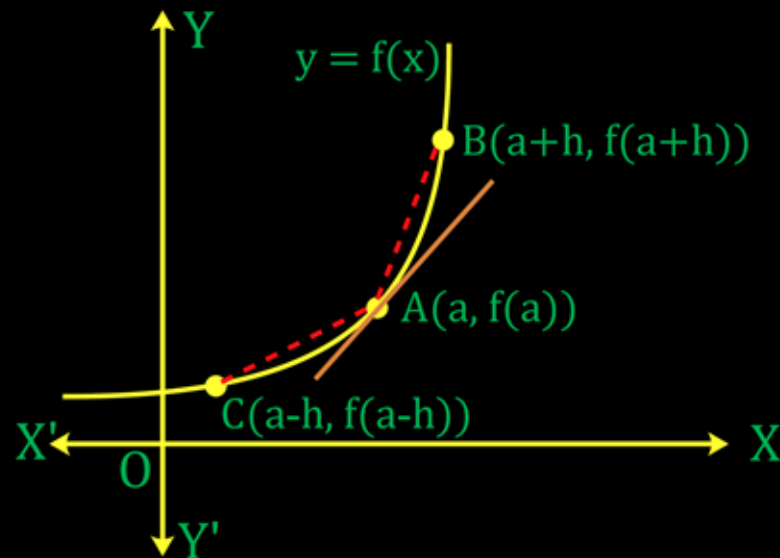
$$\text{Tangent at Point A} = \lim_{h \rightarrow 0} \frac{f(a-h)-f(a)}{-h}$$

$$\text{Slope of AB} = \frac{f(a+h)-f(a)}{a+h-a} = \frac{f(a+h)-f(a)}{h}$$

as $h \rightarrow 0 \Rightarrow B \rightarrow A$

$$\text{Tangent at Point A} = \lim_{h \rightarrow 0} \frac{f(a+h)-f(a)}{h}$$

$$\lim_{h \rightarrow 0} \frac{f(a-h)-f(a)}{-h} = \lim_{h \rightarrow 0} \frac{f(a+h)-f(a)}{h}$$



$$\lim_{h \rightarrow 0} \frac{f(a-h)-f(a)}{-h}$$

Left Hand Derivative
(LHD)

$$\lim_{h \rightarrow 0} \frac{f(a+h)-f(a)}{h}$$

Right Hand Derivative
(RHD)

Differentiability and Derivatives of Composite Functions

The derivative of a real function	f is a real function and c is a point in its domain, then the derivative of f at $x = c$ is defined by $\lim_{h \rightarrow 0} \frac{f(c + h) - f(c)}{h}$ provided this limit exists.
Derivative of f at c is denoted by $f'(c)$ or $\frac{d}{dx}(f(x)) _c$	$f'(x) = \lim_{h \rightarrow 0} \frac{f(c + h) - f(c)}{h}$ wherever the limit exists is defined to be the derivative of f .
The process of finding derivative of a function is called Differentiation .	
Differentiate $f(x)$ with respect to x to mean find $f'(x)$.	

Differentiability and Derivatives of Composite Functions

Algebra of Derivatives

- 1. $(u \pm v)' = u' \pm v'$
- 2. $(uv)' = u'v + uv'$ (Product Rule)
- 3. $\left(\frac{u}{v}\right)' = \frac{(u'v - uv')}{v^2}$, wherever $v \neq 0$ (Quotient Rule)

Derivatives of Some Standard Functions							
$f(x)$	x^n	$\sin x$	$\cos x$	$\tan x$	$\cot x$	$\sec x$	$\operatorname{cosec} x$
$f'(x)$	nx^{n-1}	$\cos x$	$-\sin x$	$\sec^2 x$	$-\operatorname{cosec}^2 x$	$\sec x \tan x$	$-\cot x \operatorname{cosec} x$

Points to Remember

If a function is

Differentiable $\rightarrow$ Definitely Continuous

Continuous $\rightarrow$ Not Necessarily Differentiable

Discontinuous $\rightarrow$ Definitely Not Differentiable

Summary

Differentiability of Standard Functions

All of the standard functions are differentiable except at certain points, as follows:

1. Polynomial functions are differentiable in its domain($\mathbb{R}$).

1. A rational function $\frac{p(x)}{q(x)}$ is differentiable **except where $q(x) = 0$** , where the function grows to infinity.

E.g. $\frac{1}{x}$ and $\frac{1}{x^2}$ both functions are **not differentiable at $x = 0$** .

1. Sines, cosines and exponents are differentiable everywhere

Tangents and **secants** are **not differentiable** at values where they are **not defined**, i.e.,

$$x = (2n + 1)\frac{\pi}{2}, n \in \mathbb{Z}$$

Cotangents and **cosecants** are **not differentiable** at values where they are **not defined**, i.e.,

$$x = n\pi, n \in \mathbb{Z}$$

Summary

If the given expression is implicit function

1. Directly differentiate with respect to x
2. Separate like and unlike terms
3. Solve for dy/dx

Derivatives of Inverse Trigonometric Function

$f(x)$	$f'(x)$	$f(x)$	$f'(x)$
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$	$\cos^{-1} x$	$-\frac{1}{\sqrt{1-x^2}}$
$\tan^{-1} x$	$\frac{1}{1+x^2}$	$\cot^{-1} x$	$-\frac{1}{1+x^2}$
$\sec^{-1} x$	$\frac{1}{x\sqrt{1-x^2}}$	$\operatorname{cosec}^{-1} x$	$-\frac{1}{x\sqrt{1-x^2}}$

Exponential and Logarithmic Functions

Exponential Function

$$y = b^x, b > 1$$

$$b = 10 \Rightarrow y = 10^x \rightarrow \text{Common Exponential Function}$$

$$b = e \Rightarrow y = e^x \rightarrow \text{Natural Exponential Function}$$

$$e \approx 2.71828$$

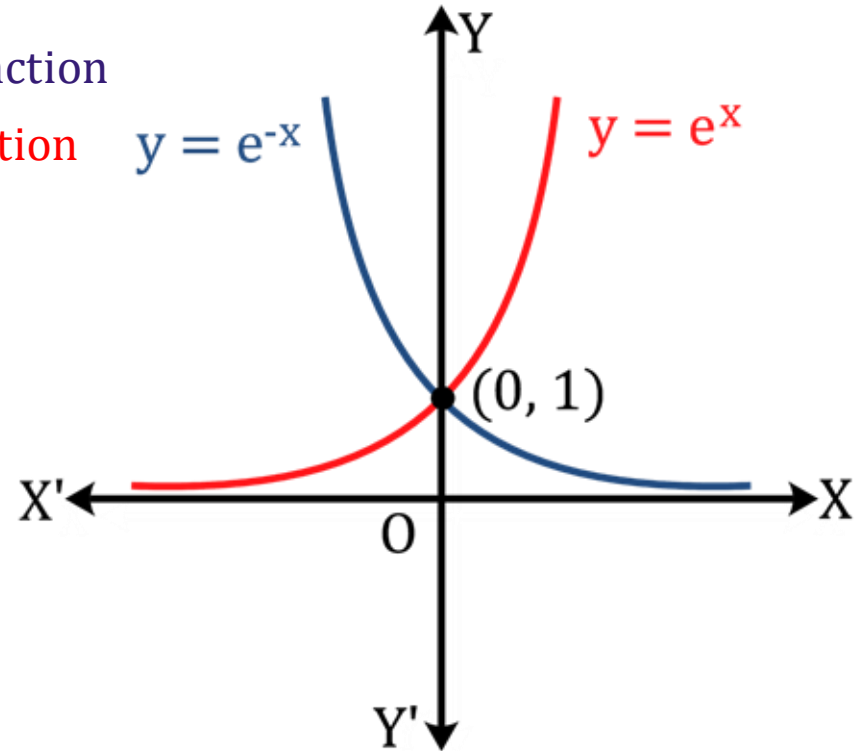
Nature

$$y = e^x \rightarrow \text{Increasing}$$

$$y = e^{-x} \rightarrow \text{Decreasing}$$

$$\text{Domain : } \mathbb{R}$$

$$\text{Range : } \mathbb{R}^+$$



Exponential and Logarithmic Functions

Logarithmic Function

$$b^y = x \Rightarrow y = \log_b x$$

$$b = 10 \Rightarrow y = \log_{10} x \rightarrow \text{Common Logarithmic Function}$$

$$b = e \Rightarrow y = \log_e x \rightarrow \text{Natural Logarithmic Function}$$

$$\log x = \log_e x = \ln x$$

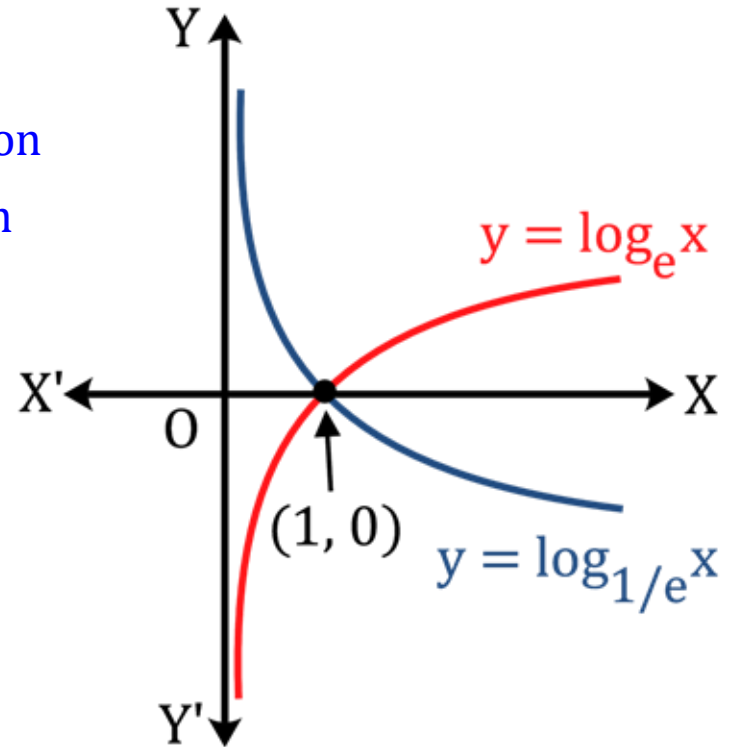
Nature

$$b > 1 \rightarrow \text{Increasing}$$

$$0 < b < 1 \rightarrow \text{Decreasing}$$

$$\text{Domain : } \mathbb{R}^+$$

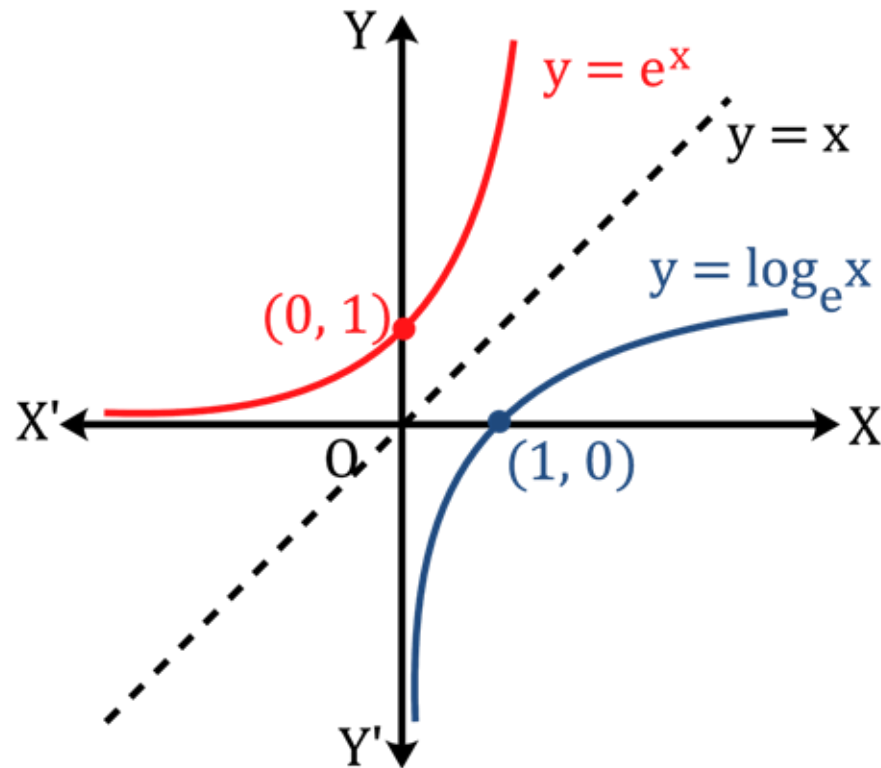
$$\text{Range : } \mathbb{R}$$



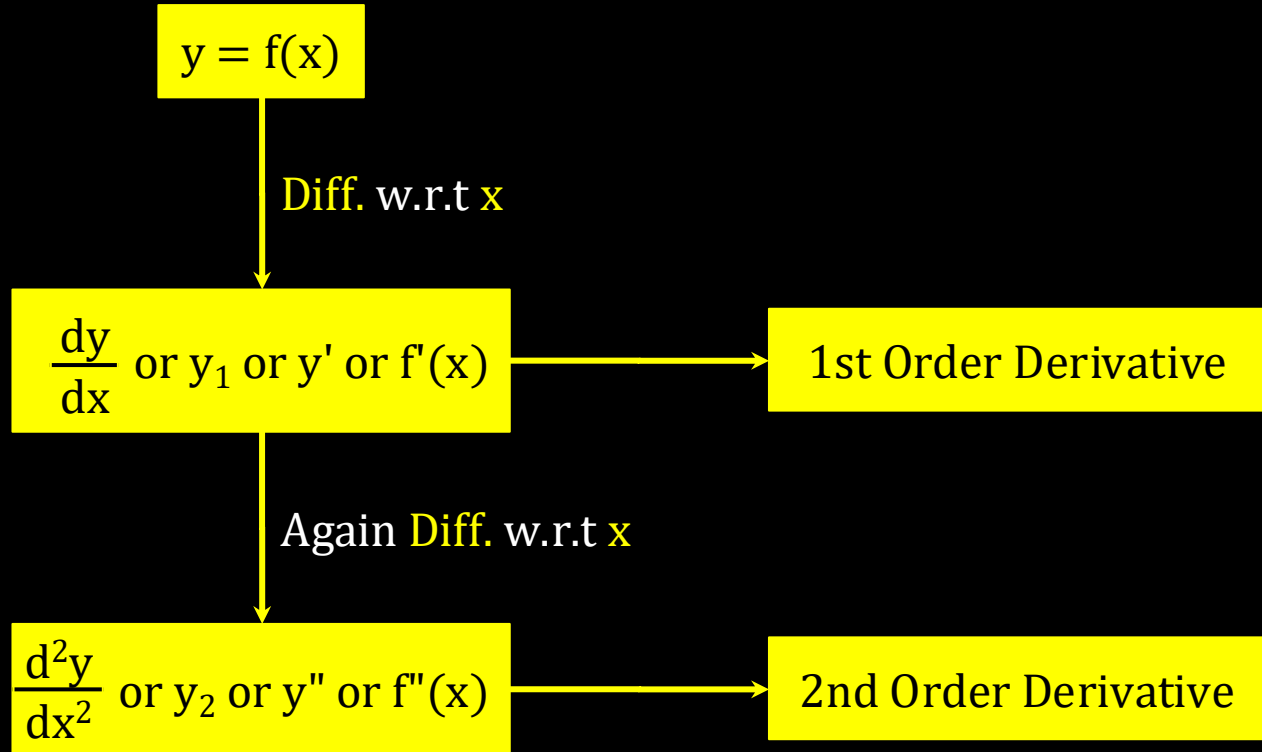
Exponential and Logarithmic Functions

Some Important Points:

1. $\log pq = \log p + \log q$
2. $\log p^n = n \log p$
1. $\log \frac{p}{q} = \log p - \log q$
1. $\log_a p = \frac{\log_e p}{\log_e a}$
1. $y = e^x$ and $y = \ln x$ are mirror images of each other with respect to $y = x$.



Second Order Derivative



Example 1 : Second Order Derivative

Example: Find $\frac{d^2y}{dx^2}$ if $y = e^x \sin 5x$.

Diff. w.r.t x

Solution: $\frac{d}{dx}(y) = \frac{d}{dx}(e^x \sin 5x) = e^x \sin 5x + 5e^x \cos 5x$

$$\Rightarrow \frac{dy}{dx} = e^x(\sin 5x + 5 \cos 5x)$$

Again diff. w.r.t x

$$\Rightarrow \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}(e^x(\sin 5x + 5 \cos 5x))$$

Product Rule

$$\Rightarrow \frac{d^2y}{dx^2} = (\sin 5x + 5 \cos 5x) \frac{d}{dx}(e^x) + e^x \frac{d}{dx}(\sin 5x + 5 \cos 5x)$$

$$\Rightarrow \frac{d^2y}{dx^2} = e^x(\sin 5x + 5 \cos 5x) + e^x(5 \cos 5x - 25 \sin 5x)$$

Simplification

$$\Rightarrow \frac{d^2y}{dx^2} = 2e^x(5 \cos 5x - 12 \sin 5x)$$

Example 2 : Second Order Derivative – Method 1

Question: If $y = \cos^{-1} x$, find $\frac{d^2y}{dx^2}$ in term of y alone.

Solution: $\frac{dy}{dx} = \frac{d}{dx} (\cos^{-1} x)$

Diff. w.r.t x

$$\Rightarrow \frac{dy}{dx} = -\frac{1}{\sqrt{1-x^2}}$$

Diff. w.r.t x

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{d}{dx} \left(-\frac{1}{\sqrt{1-x^2}} \right) = -\frac{d}{dx} ((1-x^2)^{-1/2})$$

$$\Rightarrow \frac{d^2y}{dx^2} = -\left(-\frac{1}{2} \cdot (1-x^2)^{-3/2} \cdot -2x \right)$$

Chain Rule

$$\Rightarrow \frac{d^2y}{dx^2} = -\frac{x}{(1-x^2)^{3/2}}$$

$$\because y = \cos^{-1} x \Rightarrow x = \cos y$$

$$\Rightarrow \frac{d^2y}{dx^2} = -\frac{\cos y}{(1-\cos^2 y)^{3/2}} = -\frac{\cos y}{(\sin y)^{3/2}} = -\frac{\cos y}{\sin^3 y} = -\cot y \operatorname{cosec}^2 y$$

Example 2 : Second Order Derivative – Method 2

Question: If $y = \cos^{-1} x$, find $\frac{d^2y}{dx^2}$ in term of y alone.

Solution: $\cos y = x$

Diff. w.r.t x

$$\frac{d}{dx}(\cos y) = \frac{d}{dx}(x)$$

Chain Rule

$$\Rightarrow -\sin y \cdot \frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = -\frac{1}{\sin y}$$

$$\Rightarrow \frac{dy}{dx} = -\operatorname{cosec} y$$

Diff. w.r.t x

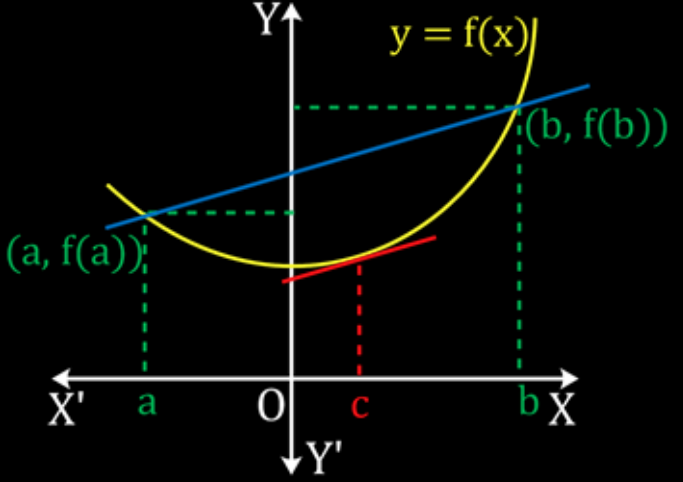
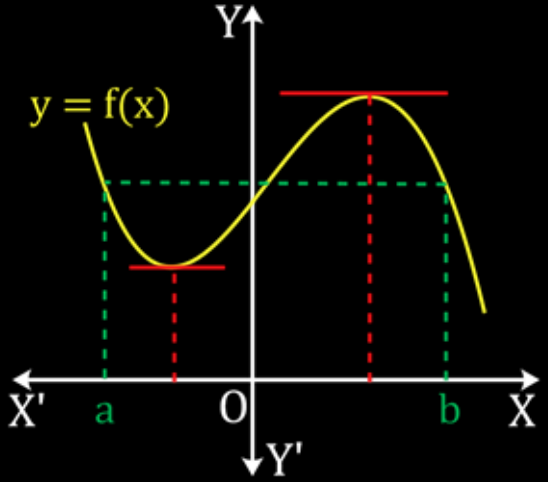
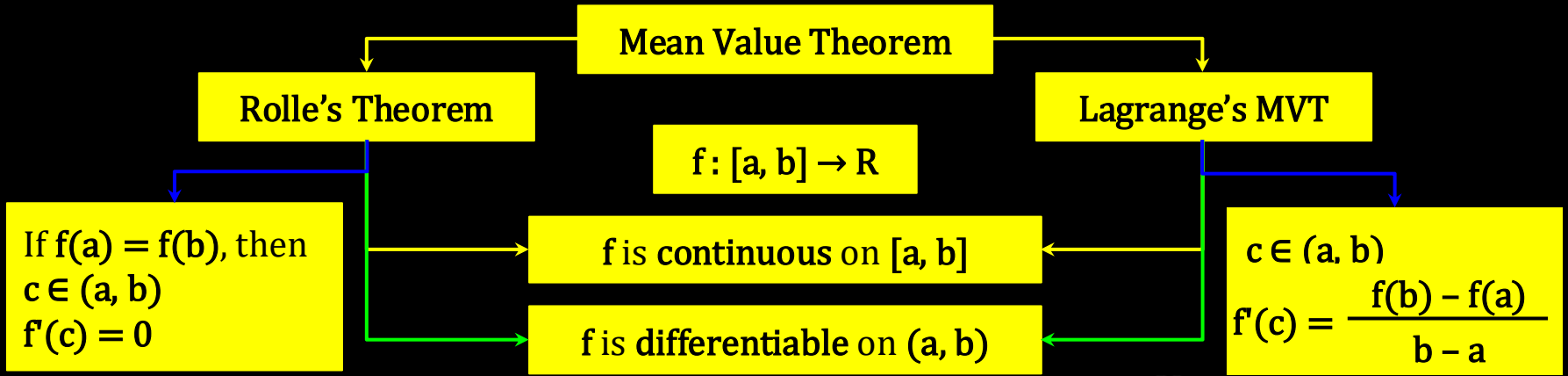
$$\Rightarrow \frac{d^2y}{dx^2} = -\frac{d}{dx}(\operatorname{cosec} y) = -\left(-\cot y \operatorname{cosec} y \frac{dy}{dx}\right)$$

Chain Rule

$$\Rightarrow \frac{d^2y}{dx^2} = \cot y \operatorname{cosec} y \cdot (-\operatorname{cosec} y) = -\cot y \operatorname{cosec}^2 y$$

Simplification

Mean Value Theorem



Extended Mean Value Theorem (Cauchy MVT)

Cauchy's Mean Value Theorem

$f: [a, b] \rightarrow \mathbb{R}$

$g: [a, b] \rightarrow \mathbb{R}$

f is continuous on $[a, b]$

f is differentiable on (a, b)

g is continuous on $[a, b]$

g is differentiable on (a, b)

If $g'(x) \neq 0$ for all x , then there exists at least one $c \in (a, b)$ such that:

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

Mean Value Theorem

Example: Verify Rolle's theorem for the function $f(x) = x^2 + 2x - 8, x \in [-4, 2]$.

Solution: $f(-4) = (-4)^2 + 2(-4) - 8 = 0$

$$f(2) = (2)^2 + 2(2) - 8 = 0$$

$$\therefore f(-4) = f(2) = 0$$

$\therefore$ From Rolle's Theorem

$$c \in (-4, 2) \text{ such that } f'(c) = 0$$

$$\therefore f(x) = x^2 + 2x - 8$$

$$\therefore f'(x) = 2x + 2$$

$$\therefore f'(c) = 2c + 2 = 0$$

$$\Rightarrow c = -1 \in (-4, 2)$$

Polynomial Function

Continuous on $[-4, 2]$

Differentiable on $(-4, 2)$

Mean Value Theorem

Question: Verify **Mean Value Theorem**, if $f(x) = x^3 - 5x^2 - 3x$ in the interval $[a, b]$, where $a = 1$ and $b = 3$. Find all $c \in (1, 3)$ for which $f'(c) = 0$.

Solution: $f(1) = (1)^3 - 5(1)^2 - 3(1) = -7$

$$f(3) = (3)^3 - 5(3)^2 - 3(3) = -27$$

$$\frac{f(b) - f(a)}{b - a} = \frac{f(3) - f(1)}{3 - 1} = \frac{-27 - (-7)}{2} = -10$$

From **Mean Value Theorem**

$$f'(c) = -10$$

$$\Rightarrow 3c^2 - 10c - 3 = -10$$

$$\Rightarrow 3c^2 - 10c + 7 = 0$$

$$\Rightarrow (c - 1)(3c - 7) = 0$$

$$\Rightarrow c = 1, 7/3$$

Factorisation

$c = 7/3 \in (1, 3)$ is the only point for which $f'(c) = 0$.

Polynomial Function
Continuous on $[1, 3]$
Differentiable on $(1, 3)$

$c \in (1, 3)$ such that

$$f(x) = x^3 - 5x^2 - 3x$$
$$\Rightarrow f'(x) = 3x^2 - 10x - 3$$

Mean Value Theorem

Question: If $f: [-5, 5] \rightarrow \mathbb{R}$ is a differentiable function and $f'(x)$ does not vanish anywhere, then prove that $f(5) \neq f(-5)$.

Solution: By Mean Value Theorem

$c \in (-5, 5)$ such that

$$f'(c) = \frac{f(5) - f(-5)}{5 - (-5)}$$

$$\Rightarrow 10f'(c) = f(5) - f(-5)$$

$$\therefore 10f'(c) \neq 0$$

$$\Rightarrow f(5) - f(-5) \neq 0$$

$$\Rightarrow f(5) \neq f(-5)$$

$$f'(c) \neq 0$$

Continuous on $[-5, 5]$

Differentiable on $(-5, 5)$

Summary

